

- Nelson Elhage, Tristan Hume, Catherine Olsson, Nicholas Schiefer, Tom Henighan, Shauna Kravec, Zac Hatfield-Dodds, Robert Lasenby, Dawn Drain, Carol Chen, Roger Grosse, Sam McCandlish, Jared Kaplan, Dario Amodei, Martin Wattenberg, and Christopher Olah. 2022. [Toy models of superposition](#). *Transformer Circuits Thread*.
- Nelson Elhage, Neel Nanda, Catherine Olsson, Tom Henighan, Nicholas Joseph, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Nova DasSarma, Dawn Drain, Deep Ganguli, Zac Hatfield-Dodds, Danny Hernandez, Andy Jones, Jackson Kernion, Liane Lovitt, Kamal Ndousse, Dario Amodei, Tom Brown, Jack Clark, Jared Kaplan, Sam McCandlish, and Chris Olah. 2021. [A mathematical framework for transformer circuits](#). *Transformer Circuits Thread*.
- Allyson Ettinger, Ahmed Elgohary, and Philip Resnik. 2016. [Probing for semantic evidence of composition by means of simple classification tasks](#). In *Proceedings of the 1st Workshop on Evaluating Vector-Space Representations for NLP*, pages 134–139, Berlin, Germany. Association for Computational Linguistics.
- Richard Futrell, Ethan Wilcox, Takashi Morita, Peng Qian, Miguel Ballesteros, and Roger Levy. 2019. [Neural language models as psycholinguistic subjects: Representations of syntactic state](#). In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 32–42, Minneapolis, Minnesota. Association for Computational Linguistics.
- Jon Gauthier, Jennifer Hu, Ethan Wilcox, Peng Qian, and Roger Levy. 2020. [SyntaxGym: An online platform for targeted evaluation of language models](#). In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics: System Demonstrations*, pages 70–76, Online. Association for Computational Linguistics.
- Atticus Geiger, Hanson Lu, Thomas Icard, and Christopher Potts. 2021. [Causal abstractions of neural networks](#). In *Advances in Neural Information Processing Systems*, volume 34, pages 9574–9586. Curran Associates, Inc.
- Atticus Geiger, Chris Potts, and Thomas Icard. 2023a. [Causal abstraction for faithful model interpretation](#). *arxiv:2301.04709*.
- Atticus Geiger, Zhengxuan Wu, Hanson Lu, Josh Rozner, Elisa Kreiss, Thomas Icard, Noah D. Goodman, and Christopher Potts. 2022. [Inducing causal structure for interpretable neural networks](#). In *International Conference on Machine Learning, ICML 2022*, volume 162 of *Proceedings of Machine Learning Research*, pages 7324–7338, Baltimore, Maryland, USA. PMLR.
- Atticus Geiger, Zhengxuan Wu, Christopher Potts, Thomas Icard, and Noah D. Goodman. 2023b. [Finding alignments between interpretable causal variables and distributed neural representations](#). *arXiv:2303.02536*.
- Nicholas Goldowsky-Dill, Chris MacLeod, Lucas Sato, and Aryaman Arora. 2023. [Localizing model behavior with path patching](#). *arXiv:2304.05969*.
- Adam Goodkind and Klinton Bicknell. 2018. [Predictive power of word surprisal for reading times is a linear function of language model quality](#). In *Proceedings of the 8th Workshop on Cognitive Modeling and Computational Linguistics (CMCL 2018)*, pages 10–18, Salt Lake City, Utah. Association for Computational Linguistics.
- Clément Guerner, Anej Svete, Tianyu Liu, Alexander Warstadt, and Ryan Cotterell. 2023. [A geometric notion of causal probing](#). *arXiv:2307.15054*.
- Kristina Gulordava, Piotr Bojanowski, Edouard Grave, Tal Linzen, and Marco Baroni. 2018. [Colorless green recurrent networks dream hierarchically](#). In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers)*, pages 1195–1205, New Orleans, Louisiana. Association for Computational Linguistics.
- Michael Hanna, Yonatan Belinkov, and Sandro Pezzelle. 2023. [When language models fall in love: Animacy processing in transformer language models](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 12120–12135, Singapore. Association for Computational Linguistics.
- Sophie Hao and Tal Linzen. 2023. [Verb conjugation in transformers is determined by linear encodings of subject number](#). In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 4531–4539, Singapore. Association for Computational Linguistics.
- John Hewitt and Percy Liang. 2019. [Designing and interpreting probes with control tasks](#). In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 2733–2743, Hong Kong, China. Association for Computational Linguistics.
- Jennifer Hu, Jon Gauthier, Peng Qian, Ethan Wilcox, and Roger Levy. 2020. [A systematic assessment of syntactic generalization in neural language models](#). In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 1725–1744, Online. Association for Computational Linguistics.
- Jennifer Hu, Kyle Mahowald, Gary Lupyan, Anna Ivanova, and Roger Levy. 2024. [Language models align with human judgments on key grammatical constructions](#). *arXiv:2402.01676*.

- Julie Kallini, Isabel Papadimitriou, Richard Futrell, Kyle Mahowald, and Christopher Potts. 2024. [Mission: Impossible language models](#). *arXiv:2401.06416*.
- Diederik P. Kingma and Jimmy Ba. 2015. [Adam: A method for stochastic optimization](#). In *3rd International Conference on Learning Representations, ICLR 2015*, San Diego, CA, USA.
- Karim Lasri, Tiago Pimentel, Alessandro Lenci, Thierry Poibeau, and Ryan Cotterell. 2022. [Probing for the usage of grammatical number](#). In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 8818–8831, Dublin, Ireland. Association for Computational Linguistics.
- Kenneth Li, Oam Patel, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. 2023. [Inference-time intervention: Eliciting truthful answers from a language model](#). In *Advances in Neural Information Processing Systems*, volume 36.
- Tal Linzen, Emmanuel Dupoux, and Yoav Goldberg. 2016. [Assessing the ability of LSTMs to learn syntax-sensitive dependencies](#). *Transactions of the Association for Computational Linguistics*, 4:521–535.
- Samuel Marks and Max Tegmark. 2023. [The geometry of truth: Emergent linear structure in large language model representations of true/false datasets](#). *arXiv:2310.06824*.
- Rebecca Marvin and Tal Linzen. 2018. [Targeted syntactic evaluation of language models](#). In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 1192–1202, Brussels, Belgium. Association for Computational Linguistics.
- Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. 2022. [Locating and editing factual associations in GPT](#). In *Advances in Neural Information Processing Systems*, volume 35, pages 17359–17372. Curran Associates, Inc.
- Tomas Mikolov, Wen-tau Yih, and Geoffrey Zweig. 2013. [Linguistic regularities in continuous space word representations](#). In *Proceedings of the 2013 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 746–751, Atlanta, Georgia. Association for Computational Linguistics.
- Neel Nanda, Andrew Lee, and Martin Wattenberg. 2023. [Emergent linear representations in world models of self-supervised sequence models](#). In *Proceedings of the 6th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP*, pages 16–30, Singapore. Association for Computational Linguistics.
- Chris Olah. 2022. [Mechanistic interpretability, variables, and the importance of interpretable bases](#). *Transformer Circuits Thread*.
- Kiho Park, Yo Joong Choe, and Victor Veitch. 2023. [The linear representation hypothesis and the geometry of large language models](#). *arXiv:2311.03658*.
- Judea Pearl. 2009. *Causality: Models, Reasoning, and Inference*, 2nd edition. Cambridge University Press.
- Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake VanderPlas, Alexandre Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot, and Edouard Duchesnay. 2011. [Scikit-learn: Machine learning in Python](#). *The Journal of Machine Learning Research*, 12:2825–2830.
- Cory Shain, Clara Meister, Tiago Pimentel, Ryan Cotterell, and Roger Levy. 2024. [Large-scale evidence for logarithmic effects of word predictability on reading time](#). *Proceedings of the National Academy of Sciences*. To appear.
- Nathaniel J. Smith and Roger Levy. 2013. [The effect of word predictability on reading time is logarithmic](#). *Cognition*, 128(3):302–319.
- Curt Tigges, Oskar John Hollinsworth, Atticus Geiger, and Neel Nanda. 2023. [Linear representations of sentiment in large language models](#). *arXiv:2310.15154*.
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. [Attention is all you need](#). In I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 30, pages 5998–6008. Curran Associates, Inc.
- Jesse Vig, Sebastian Gehrmann, Yonatan Belinkov, Sharon Qian, Daniel Nevo, Yaron Singer, and Stuart Shieber. 2020. [Investigating gender bias in language models using causal mediation analysis](#). In *Advances in Neural Information Processing Systems*, volume 33, pages 12388–12401. Curran Associates, Inc.
- Kevin Ro Wang, Alexandre Variengien, Arthur Conmy, Buck Shlegeris, and Jacob Steinhardt. 2023. [Interpretability in the wild: a circuit for indirect object identification in GPT-2 small](#). In *The Eleventh International Conference on Learning Representations, ICLR 2023*, Kigali, Rwanda.
- Alex Warstadt and Samuel R. Bowman. 2022. [What artificial neural networks can tell us about human language acquisition](#). *Algebraic Structures in Natural Language*, pages 17–60.
- Alex Warstadt, Alicia Parrish, Haokun Liu, Anhad Mohananey, Wei Peng, Sheng-Fu Wang, and Samuel R. Bowman. 2020. [BLiMP: The benchmark of linguistic minimal pairs for English](#). *Transactions of the Association for Computational Linguistics*, 8:377–392.

Ethan Wilcox, Clara Meister, Ryan Cotterell, and Tiago Pimentel. 2023a. [Language model quality correlates with psychometric predictive power in multiple languages](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 7503–7511, Singapore. Association for Computational Linguistics.

Ethan Gottlieb Wilcox, Richard Futrell, and Roger Levy. 2023b. [Using computational models to test syntactic learnability](#). *Linguistic Inquiry*, pages 1–44.

Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, Tim Rault, Remi Louf, Morgan Funtowicz, Joe Davison, Sam Shleifer, Patrick von Platen, Clara Ma, Yacine Jernite, Julien Plu, Canwen Xu, Teven Le Scao, Sylvain Gugger, Mariama Drame, Quentin Lhoest, and Alexander Rush. 2020. [Transformers: State-of-the-art natural language processing](#). In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 38–45, Online. Association for Computational Linguistics.

Zhengxuan Wu, Atticus Geiger, Aryaman Arora, Jing Huang, Noah D. Goodman, Christopher D. Manning, and Christopher Potts. 2024. [pyvene: A library for understanding and improving PyTorch models via interventions](#). Under review.

Zhengxuan Wu, Atticus Geiger, Christopher Potts, and Noah D. Goodman. 2023. [Interpretability at scale: Identifying causal mechanisms in Alpaca](#). In *Advances in Neural Information Processing Systems*, volume 36.

Takateru Yamakoshi, James McClelland, Adele Goldberg, and Robert Hawkins. 2023. [Causal interventions expose implicit situation models for common-sense language understanding](#). In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 13265–13293, Toronto, Canada. Association for Computational Linguistics.